

electronics that can adhere seamlessly to human skin or within the body are highly desirable for applications such as health monitoring, medical treatment, medical implants, and biological studies and for technologies that include human-machine interfaces, soft robotics, and augmented reality.

Rendering such electronics that are soft and stretchable like human skin would make them more comfortable to wear. Structural engineering of rigid inorganic and organic devices has enabled circuit-level stretchability, but this requires sophisticated fabrication techniques and usually suffers from reduced densities of devices within an array.

Scientists have developed a super-flexible, super-durable, and super-sensitive material that can be the basis for future synthetic skin. It contains organic transistors and a layer of elastic, letting it stretch without any damage. Further, it can also be self-powered as it can contain a series of elastic solar cells.

Artificial eyes

Medical science has been able to restore hearing, but restoring sight to the blind is much more complicated. When people lose their sight, their retinas no longer send signals from their photoreceptors to their brains. To make an artificial eye, it is important to understand how the retina processes those signals, and scientists haven't been able to crack this yet.

The artificial eye produces artificial retinas, whose chips convert images into electronic signals, and tiny projectors convert electronic signals into light. These artificial eyes have indeed restored sight to blind mice, and the follow-up experiments on monkeys offer hope for trials on humans.

The most successful function-replacing artificial eye so far is an external miniature digital camera with a remote unidirectional electronic interface implanted on the retina, optic nerve, or other related locations inside the brain. The present state-of-the-art yields only partial functionality, such as recognising levels of brightness, swatches of colour, and/or basic geometric shapes. No doubt, the problem of creating a completely functional artificial electronic eye is

even more complex but advances towards tackling the complexity of the artificial connection to the retina, optic nerve, or related brain areas, combined with ongoing advances in computer science, are expected to dramatically improve the performance of this technology.

Printed ears

In cases when a person is profoundly deaf or severely hard of hearing in both ears, a cochlear implant may be surgically implanted. Cochlear implants bypass most of the peripheral auditory system to provide a sense of sound via a microphone and some electronics that reside outside the skin, generally behind the ear.

The external components transmit a signal to an array of electrodes placed in the cochlea, which in turn stimulates the cochlear nerve, but internal implants do nothing for the visible parts of the ear. Traditional replacements to ears look and feel like plastic toys. But recently, researchers have come up with a new method that makes flexible, realistic ears out of real cells. The artificial ear then just needs a few days to grow in nutrients before it's ready to be implanted in a subject. Artificial ears will be a huge benefit to those who suffer injuries.

E-nose

Smells are composed of molecules, and each of these molecules has a corresponding sized and shaped receptor in the human nose. When a specific receptor receives a molecule, it sends a signal to brain. Brain identifies smell associated with a particular molecule.

An electronic nose is a device that detects smell more effectively than



Fig. 5: Artificial eye



Fig. 6: Printed ear

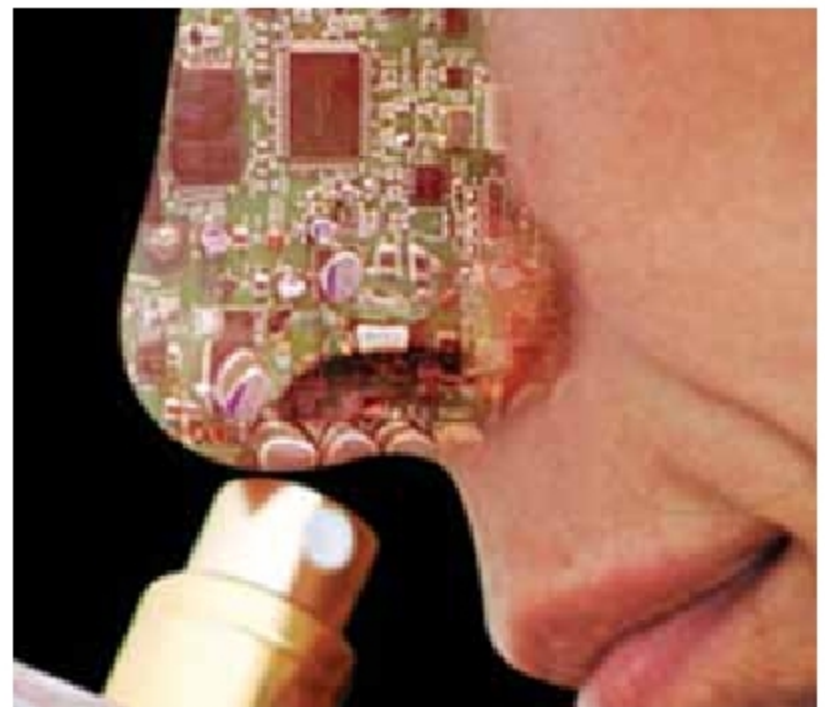


Fig. 7: e-Nose

a human nose. An electronic nose consists of a mechanism for chemical detection. It is an intelligent sensing device that uses an array of gas sensors that are overlapping selectively along with a patterned reorganisation component.

Electronic noses work in a similar manner as a human nose. The electronic nose uses sensors as the receptor. When a specific sensor receives the molecules, it transmits the signal to a program for processing rather than to brain.

Nowadays, electronic noses provide several benefits to a variety of com-